



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

NDIC - DATEX II Predefined Location Publication - Predefined Location Set

Release 1.1rev0

Národní Dopravní Informační Centrum

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The specification defines format for publishing predefined location data used to reference status oriented events (travel time, weather station reports, parking, tunnels, etc. accidents etc.).

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NTIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NTIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DDR

data distribution interface, a proprietary format used by NDIC

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [\[d2ref\]](#). For additional information see [\[d2supp\]](#).

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [16157-1]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [16157-3], *ElaboratedDataPublication* [16157-5], *PredefinedLocationsPublication* [16157-3] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing of predefined locations used for referencing of status information (travel times, level of service, tunnel status, parking occupancy, weather station reports etc.).

2.1 Publication type *PredefinedLocationPublication*

Format is based on DATEX II publication of type *PredefinedLocationPublication* [16157-3]. The profile uses most of data structures, provided by this type of publication and extends it by proprietary location referencing.

```

1  <?xml version="1.0" encoding="utf-8"?>
2  <d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/
   → XMLSchema-instance" xmlns="http://datex2.eu/schema/2/2_0"
   → xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
3    <exchange>
4      <supplierIdentification>
5        <country>cz</country>
6        <nationalIdentifier>NDIC</nationalIdentifier>
7      </supplierIdentification>
8    </exchange>
9    <payloadPublication xsi:type="PredefinedLocationsPublication" lang="cs">
10     <feedType>x-format:cz-ndic_d2-predefined-location-set-v1.1</feedType>
11     <publicationTime>2017-08-15T10:09:22+02:00</publicationTime>
12     <publicationCreator>
13       <country>cz</country>
14       <nationalIdentifier>NDIC</nationalIdentifier>
15     </publicationCreator>
16     <headerInformation>
17       <confidentiality>noRestriction</confidentiality>
18       <informationStatus>real</informationStatus>
19     </headerInformation>

```

2.2 Predefined location description (point or linear segment)

Location is a point or linear segment on road network. Location type area is not used. The locations are described by means of several methods, each *predefinedLocationContainer* describes one location (by one or more methods). Element *predefinedLocationContainer* is unanimously identified by unique identifier and version.

Each *predefinedLocationContainer* could contain a description in an element *PredefinedLocationName*.

```

82 <predefinedLocationContainer xsi:type="PredefinedLocation" id="da0a5990-
   → 530c-426a-a6fd-ce88aeeb8aab" version="1">
83   <predefinedLocationName>

```

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```

84     <values>
85       <value lang="cs">P+R Chodov, D1 0 km</value>
86     </values>
87   </predefinedLocationName>
88   <location xsi:type="Linear">
89     <alertCLinear xsi:type="AlertCMethod2Linear">
90       <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
91       <alertCLocationTableNumber>25</alertCLocationTableNumber>
92       <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
93       <alertCDirection>
94         <alertCDirectionCoded>positive</alertCDirectionCoded>
95       </alertCDirection>
96       <alertCMethod2PrimaryPointLocation>
97         <alertCLocation>
98           <specificLocation>23078</specificLocation>
99         </alertCLocation>
100       </alertCMethod2PrimaryPointLocation>
101       <alertCMethod2SecondaryPointLocation>
102         <alertCLocation>
103           <specificLocation>23078</specificLocation>
104         </alertCLocation>
105       </alertCMethod2SecondaryPointLocation>
106     </alertCLinear>
107   <globalNetworkLinear>
108     <networkVersion>
109       <values>
110         <value lang="cs">16.12</value>
111       </values>
112     </networkVersion>
113     <linearGeometryType>continuous</linearGeometryType>
114     <startPoint>
115       <sjtskPointCoordinates>
116         <sjtskX>-738706</sjtskX>
117         <sjtskY>-1049716</sjtskY>
118       </sjtskPointCoordinates>
119     </startPoint>
120     <endPoint>
121       <sjtskPointCoordinates>
122         <sjtskX>-738628</sjtskX>
123         <sjtskY>-1049838</sjtskY>
124       </sjtskPointCoordinates>
125     </endPoint>
126     <linearWithinLinearGNElement>
127       <sectionId>822309</sectionId>
128       <directionRelativeOnLinearSection>aligned</
129   ↪ directionRelativeOnLinearSection>
130       <orderOfSection>0</orderOfSection>
131       <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
132         <percentageDistanceAlong>0.857295694891544</percentageDistanceAlong>
133       </fromPoint>
134       <toPoint xsi:type="PercentageDistanceAlongLinearElement">
135         <percentageDistanceAlong>1</percentageDistanceAlong>
136       </toPoint>
137     </linearWithinLinearGNElement>
138     <linearWithinLinearGNElement>
139       <sectionId>822310</sectionId>

```

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```

139     <directionRelativeOnLinearSection>aligned</
→ directionRelativeOnLinearSection>
140     <orderOfSection>1</orderOfSection>
141     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
142       <percentageDistanceAlong>0</percentageDistanceAlong>
143     </fromPoint>
144     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
145       <percentageDistanceAlong>1</percentageDistanceAlong>
146     </toPoint>
147   </linearWithinLinearGNElement>
148   <linearWithinLinearGNElement>
149     <sectionId>822311</sectionId>
150     <directionRelativeOnLinearSection>aligned</
→ directionRelativeOnLinearSection>
151     <orderOfSection>2</orderOfSection>
152     <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
153       <percentageDistanceAlong>0</percentageDistanceAlong>
154     </fromPoint>
155     <toPoint xsi:type="PercentageDistanceAlongLinearElement">
156       <percentageDistanceAlong>0.662356475026882</percentageDistanceAlong>
157     </toPoint>
158   </linearWithinLinearGNElement>
159 </globalNetworkLinear>
160 </location>
161 </predefinedLocationContainer>

```


TIME VALIDITY

The publication start of validity is defined by *publicationTime* for all predefined locations in the publication. No other time record is contained in the publication. In case of new publication due to changes to predefined location set, old location remains the same, new locations get new id and updated locations got its version increased.

```
10 <feedType>x-format:cz-ndic_d2-predefined-location-set-v1.1</feedType>  
11 <publicationTime>2017-08-15T10:09:22+02:00</publicationTime>
```

LOCATION REFERENCING

All locations in this publication are predefined and each is contained within element *PrefefinedLocationContainer* globally identified by its identifier and version (these are used to reference the location from other publications). Element *location* (descendant of *NetworkLocation*) with type *Point* or *Linear* contains description of one physical location in one or more location referencing methods.

Three location description methods for a linear location and four location description methods for point location can be used to describe the location. The method use depends on method availability at the location. At least one location description should be used.

4.1 Location description by means of ALERT-C

The method describes position by ALERT-C codes, by class *alertCLinear* of the *AlertCMethod2Linear* type for linear locations and by class *alertCPoint* of the *AlertCMethod2Point* type for point locations. Other methods of ALERT-C description are not used. The use of this method depends on the presence of the ALERT-C location at a given location.

The *alertCLinear* method uses the primary location (element *alertCMethod2PrimaryPointLocation*), secondary location (element *alertCMethod2SecondaryPointLocation*) and direction (*alertCDirection*) element to describe the event. Of course, the method also includes elements that allows to identify a particular location table.

```
92 <alertCLinear xsi:type="AlertCMethod2Linear">
93   <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
94   <alertCLocationTableNumber>25</alertCLocationTableNumber>
95   <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
96   <alertCDirection>
97     <alertCDirectionCoded>positive</alertCDirectionCoded>
98   </alertCDirection>
99   <alertCMethod2PrimaryPointLocation>
100     <alertCLocation>
101       <specificLocation>23078</specificLocation>
102     </alertCLocation>
103   </alertCMethod2PrimaryPointLocation>
104   <alertCMethod2SecondaryPointLocation>
105     <alertCLocation>
106       <specificLocation>23078</specificLocation>
107     </alertCLocation>
108   </alertCMethod2SecondaryPointLocation>
109 </alertCLinear>
```

Element *alertCLinear* can appear more than once in a situation location description. This could happen for long events that must be described by more than one ALERT-C route. ALERT-C allows to chain (reference between) locations only within one route.

The primary location is, by the ALERT-C principle, the place where the accident occurred or where the driver leaves the problem area. The secondary location is the end of the traffic queue or the place where the driver first

enters the problem area.

This method is not optimal because the RDS-TMC uses to describe the event location the primary location, direction and extent, looking up the secondary and intermittent locations in the location table. Therefore, if the recipient systems are adapted to RDS-TMC event reconstruction, it will be necessary to modify them.

The *alertCPoint* method is used to describe the point location (the inside of a junction or a tunnel, a short section). This method uses the primary location (element *alertCMethod2PrimaryPointLocation*) and the direction (*alertCDirection*) to describe the event. Of course, the method also includes elements that allow to identify a particular location table.

```

121 <alertCPoint xsi:type="AlertCMethod2Point">
122   <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
123   <alertCLocationTableNumber>25</alertCLocationTableNumber>
124   <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
125   <alertCDirection>
126     <alertCDirectionCoded>positive</alertCDirectionCoded>
127   </alertCDirection>
128   <alertCMethod2PrimaryPointLocation>
129     <alertCLocation>
130       <specificLocation>23078</specificLocation>
131     </alertCLocation>
132   </alertCMethod2PrimaryPointLocation>
133 </alertCPoint>

```

However, if it is necessary to express the short location covered by one ALERT-C point also by other methods, then the *alertCLinear* method has to be used (where the secondary location is equal to the primary), since the methods describing one location must be of same type e.g. point or linear.

4.2 Location description using a textual description of the road network

The method uses the “address” of the street or road (name, area, number, etc.) and the distance of the event location from the beginning and the end of the road. For description of linear locations, the *linearWithinLinearElement* class is used and for point locations the *pointAlongLinearElement* class is used. The use of this method depends on the presence of a proper description of the road (ideal for motorways).

The *roadNumber* element describes the number of the affected road, element *DirectionRelativeOnLinearSection* describes the direction of affected traffic, and the *fromPoint* and *toPoint* elements describes distance of the start and the end of the event from the beginning of the road.

```

145 <linearWithinLinearElement>
146   <directionRelativeOnLinearSection>opposite</
→ directionRelativeOnLinearSection>
147   <linearElement>
148     <roadNumber>D11</roadNumber>
149   </linearElement>
150   <fromPoint xsi:type="DistanceFromLinearElementStart">
151     <distanceAlong>8016</distanceAlong>
152   </fromPoint>
153   <toPoint xsi:type="DistanceFromLinearElementStart">
154     <distanceAlong>7942</distanceAlong>
155   </toPoint>
156 </linearWithinLinearElement>

```

The method for description of a point event *PointAlongLinearElement* differs by simply describing one distance from the beginning of the identified road instead of two.

```

168     <pointAlongLinearElement>
169       <linearElement>
170         <roadNumber>D11</roadNumber>
171       </linearElement>
172       <distanceAlongLinearElement xsi:type=
→ "DistanceFromLinearElementStart">
173         <distanceAlong>8016</distanceAlong>
174       </distanceAlongLinearElement>
175     </pointAlongLinearElement>

```

4.3 Location description using the Global Network extension

This is a DATEX II model extension type B.

The method uses for location description identifiers of affected sections of the road network, specified by Global Network (GN) database, and event start and end coordinates (in the S-JTSK reference system). The last and initial segments are truncated by the relative distance of the start and end points of the event relative to the GN segment.

The *globalNetworkLinear* class is used to describe the linear location and *pointWithinLinearGNElement* for point location description.

The *globalNetworkLinear* element describes the location of one event on the network. It contains the network version identifier *networkVersion*, the points bounding the beginning and end of the event (*startPoint* and *endPoint* in coordinates, here S-JTSK). And one or more *linearWithinLinearGNElement* elements that describe specific segments of the GN database by their direction, identifier, location order, and relative truncation (for the first and last sections).

This is native means of location referencing by NDIC.

```

28     <globalNetworkLinear>
29       <networkVersion>
30         <values>
31           <value lang="cs">16.12</value>
32         </values>
33       </networkVersion>
34       <linearGeometryType>continuous</linearGeometryType>
35       <startPoint>
36         <sjtskPointCoordinates>
37           <sjtskX>-738706</sjtskX>
38           <sjtskY>-1049716</sjtskY>
39         </sjtskPointCoordinates>
40       </startPoint>
41       <endPoint>
42         <sjtskPointCoordinates>
43           <sjtskX>-738628</sjtskX>
44           <sjtskY>-1049838</sjtskY>
45         </sjtskPointCoordinates>
46       </endPoint>
47       <linearWithinLinearGNElement>
48         <sectionId>822309</sectionId>
49         <directionRelativeOnLinearSection>aligned</
→ directionRelativeOnLinearSection>
50         <orderOfSection>0</orderOfSection>
51         <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
52           <percentageDistanceAlong>0.857295694891544</
→ percentageDistanceAlong>
53         </fromPoint>

```

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```

54      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
55        <percentageDistanceAlong>1</percentageDistanceAlong>
56      </toPoint>
57    </linearWithinLinearGNElement>
58    <linearWithinLinearGNElement>
59      <sectionId>822310</sectionId>
60      <directionRelativeOnLinearSection>aligned</
61    <directionRelativeOnLinearSection>
62      <orderOfSection>1</orderOfSection>
63      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
64        <percentageDistanceAlong>0</percentageDistanceAlong>
65      </fromPoint>
66      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
67        <percentageDistanceAlong>1</percentageDistanceAlong>
68      </toPoint>
69    </linearWithinLinearGNElement>
70    <linearWithinLinearGNElement>
71      <sectionId>822311</sectionId>
72      <directionRelativeOnLinearSection>aligned</
73    <directionRelativeOnLinearSection>
74      <orderOfSection>2</orderOfSection>
75      <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
76        <percentageDistanceAlong>0</percentageDistanceAlong>
77      </fromPoint>
78      <toPoint xsi:type="PercentageDistanceAlongLinearElement">
79        <percentageDistanceAlong>0.662356475026882</
80    <percentageDistanceAlong>
81      </toPoint>
82    </linearWithinLinearGNElement>
83  </globalNetworkLinear>

```

The method for description of a point location *pointWithinLinearGNElement* only describes one distance from the beginning of the identified segment of GN database instead of two distances. Is very similar to the *pointAlongLinearElement* method.

```

187    <pointWithinLinearGNElement>
188      <sectionId>822309</sectionId>
189      <directionRelativeOnLinearSection>aligned</
190    <directionRelativeOnLinearSection>
191      <networkVersion>
192        <values>
193          <value lang="cs">16.12</value>
194        </values>
195      </networkVersion>
196      <distanceAlongLinearElement xsi:type="
197    <"PercentageDistanceAlongLinearElement">
198      <percentageDistanceAlong>0.662356475026882</
199    <percentageDistanceAlong>
200      </distanceAlongLinearElement>
201    </pointWithinLinearGNElement>

```

4.4 Location description using Coordinates

The method uses geographic coordinates to describe a situation location. Two methods may be used based on used coordinate reference system.

Method *pointByCoordinates* is used for description of position in WGS-84 coordinate system.

```
231 <pointByCoordinates>
232   <pointCoordinates>
233     <latitude>50.16949</latitude>
234     <longitude>14.46131</longitude>
235   </pointCoordinates>
236 </pointByCoordinates>
```

Method *pointByMultiCoordinates* is used to describe location by one or more coordinate systems, one of which being Czech system S-JTSK and second WGS-84. This is a DATEX II model extension type B.

```
210 <pointByMultiCoordinates>
211   <sjtskPointCoordinates>
212     <sjtskX>-738731</sjtskX>
213     <sjtskY>-1034365</sjtskY>
214   </sjtskPointCoordinates>
215   <pointCoordinates>
216     <latitude>50.16949</latitude>
217     <longitude>14.46131</longitude>
218   </pointCoordinates>
219 </pointByMultiCoordinates>
```

APPENDIX A: PUBLICATION EXAMPLE

5.1 predefined_location_set.xml - predefined location set with two locations

Publication contains two predefined locations.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2" xmlns:xsi="http://www.w3.org/2001/
→XMLSchema-instance" xmlns="http://datex2.eu/schema/2/2_0"
→xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication xsi:type="PredefinedLocationsPublication" lang="cs">
    <feedType>x-format:cz-ndic_d2-predefined-location-set-v1.1</feedType>
    <publicationTime>2017-08-15T10:09:22+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <headerInformation>
      <confidentiality>noRestriction</confidentiality>
      <informationStatus>real</informationStatus>
    </headerInformation>
    <predefinedLocationContainer xsi:type="PredefinedLocation" id="dde51831-
→2f37-4f07-8429-286a9d08106c" version="1">
      <predefinedLocationName>
        <values>
          <value lang="cs">Podolské nábřeží: Podolská &gt; zúžení U Podolského
→sanatoria</value>
        </values>
      </predefinedLocationName>
      <location xsi:type="Linear">
        <globalNetworkLinear>
          <networkVersion>
            <values>
              <value lang="cs">15.06</value>
            </values>
          </networkVersion>
          <linearGeometryType>continuous</linearGeometryType>
          <startPoint>
```

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```

    <sjtskPointCoordinates>
      <sjtskX>-743357.999982555</sjtskX>
      <sjtskY>-1046109.99933721</sjtskY>
    </sjtskPointCoordinates>
  </startPoint>
  <endPoint>
    <sjtskPointCoordinates>
      <sjtskX>-743373.999834907</sjtskX>
      <sjtskY>-1045915.0002325</sjtskY>
    </sjtskPointCoordinates>
  </endPoint>
  <linearWithinLinearGNElement>
    <sectionId>2378839</sectionId>
    <directionRelativeOnLinearSection>aligned</
    ↳directionRelativeOnLinearSection>
    <orderOfSection>0</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0.000663000030517578</
    ↳percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>66.300003051757812</
    ↳percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
  <linearWithinLinearGNElement>
    <sectionId>2378840</sectionId>
    <directionRelativeOnLinearSection>aligned</
    ↳directionRelativeOnLinearSection>
    <orderOfSection>1</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>103.09999847412109</
    ↳percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
  <linearWithinLinearGNElement>
    <sectionId>713522</sectionId>
    <directionRelativeOnLinearSection>aligned</
    ↳directionRelativeOnLinearSection>
    <orderOfSection>2</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>28.599714279174805</
    ↳percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
</globalNetworkLinear>
</location>
</predefinedLocationContainer>
<predefinedLocationContainer xsi:type="PredefinedLocation" id="da0a5990-
    ↳530c-426a-a6fd-ce88aeeb8aab" version="1">

```

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```

<predefinedLocationName>
  <values>
    <value lang="cs">P+R Chodov, D1 0 km</value>
  </values>
</predefinedLocationName>
<location xsi:type="Linear">
  <alertCLinear xsi:type="AlertCMethod2Linear">
    <alertCLocationCountryCode>CZ</alertCLocationCountryCode>
    <alertCLocationTableNumber>25</alertCLocationTableNumber>
    <alertCLocationTableVersion>6.0</alertCLocationTableVersion>
    <alertCDirection>
      <alertCDirectionCoded>positive</alertCDirectionCoded>
    </alertCDirection>
    <alertCMethod2PrimaryPointLocation>
      <alertCLocation>
        <specificLocation>23078</specificLocation>
      </alertCLocation>
    </alertCMethod2PrimaryPointLocation>
    <alertCMethod2SecondaryPointLocation>
      <alertCLocation>
        <specificLocation>23078</specificLocation>
      </alertCLocation>
    </alertCMethod2SecondaryPointLocation>
  </alertCLinear>
<globalNetworkLinear>
  <networkVersion>
    <values>
      <value lang="cs">16.12</value>
    </values>
  </networkVersion>
  <linearGeometryType>continuous</linearGeometryType>
  <startPoint>
    <sjtskPointCoordinates>
      <sjtskX>-738706</sjtskX>
      <sjtskY>-1049716</sjtskY>
    </sjtskPointCoordinates>
  </startPoint>
  <endPoint>
    <sjtskPointCoordinates>
      <sjtskX>-738628</sjtskX>
      <sjtskY>-1049838</sjtskY>
    </sjtskPointCoordinates>
  </endPoint>
  <linearWithinLinearGNElement>
    <sectionId>822309</sectionId>
    <directionRelativeOnLinearSection>aligned</
→directionRelativeOnLinearSection>
    <orderOfSection>0</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0.857295694891544</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
</linearWithinLinearGNElement>

```

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```

    <sectionId>822310</sectionId>
    <directionRelativeOnLinearSection>aligned</
→ directionRelativeOnLinearSection>
    <orderOfSection>1</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>1</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
  <linearWithinLinearGNElement>
    <sectionId>822311</sectionId>
    <directionRelativeOnLinearSection>aligned</
→ directionRelativeOnLinearSection>
    <orderOfSection>2</orderOfSection>
    <fromPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0</percentageDistanceAlong>
    </fromPoint>
    <toPoint xsi:type="PercentageDistanceAlongLinearElement">
      <percentageDistanceAlong>0.662356475026882</percentageDistanceAlong>
    </toPoint>
  </linearWithinLinearGNElement>
</globalNetworkLinear>
</location>
</predefinedLocationContainer>
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *predefinedLocationContainer*, (in our case there are two).

Attribute *xsi:type="PredefinedLocationsPublication"* in element *payloadPublication* determines, that this envelope is of type predefined location publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of predefined locations

Predefined locations are described in elements *predefinedLocationContainer* using these unique identifiers *id="dde51831-2f37-4f07-8429-286a9d08106c"* and *id="da0a5990-530c-426a-a6fd-ce88aeeb8aab"*.

Location description (dde51831-2f37-4f07-8429-286a9d08106c)

Element *predefinedLocationName* contains a textual description of a physical location.

Element *location* of type *xsi:type="Linear"* contains individual location description methods, in our case just one the description using Global Network links in element *globalNetworkLinear*.

Location description (da0a5990-530c-426a-a6fd-ce88aeeb8aab)

Element *predefinedLocationName* contains a textual description of a physical location.

Element *location* of type *xsi:type="Linear"* contains individual location description methods, in our case two. First one the description using location tables and ALERT-C location in *alertCLinear* element, the second one using Global Network links in element *globalNetworkLinear*.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_IntermediatePointOnLinearElement">
9     <xs:sequence>
10      <xs:element name="referent" type="D2LogicalModel:Referent" minOccurs="1"
  ↳ maxOccurs="1" />
11    </xs:sequence>
12    <xs:attribute name="index" type="xs:int" use="required" />
13  </xs:complexType>
14  <xs:complexType name="_LinearExtensionType">
15    <xs:sequence>
16      <xs:element name="openlrExtendedLinear" type=
  ↳ "D2LogicalModel:OpenlrExtendedLinear" minOccurs="0" />
17      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
18    </xs:sequence>
19  </xs:complexType>
20  <xs:complexType name="_PointExtensionType">
21    <xs:sequence>
22      <xs:element name="openlrExtendedPoint" type="D2LogicalModel:OpenlrExtendedPoint
  ↳ " minOccurs="0" />
23      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
24    </xs:sequence>
25  </xs:complexType>
26  <xs:complexType name="AlertCDirection">
27    <xs:sequence>
28      <xs:element name="alertCDirectionCoded" type="D2LogicalModel:AlertCDirectionEnum
  ↳ " minOccurs="1" maxOccurs="1" />
29      <xs:element name="alertCDirectionNamed" type="D2LogicalModel:MultilingualString
  ↳ " minOccurs="0" maxOccurs="1" />

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```

30     <xs:element name="alertCDirectionSense" type="D2LogicalModel:Boolean" minOccurs=
    ↪ "0" maxOccurs="1" />
31     <xs:element name="alertCDirectionExtension" type="D2LogicalModel:_ExtensionType"
    ↪ " minOccurs="0" />
32     </xs:sequence>
33 </xs:complexType>
34 <xs:simpleType name="AlertCDirectionEnum">
35     <xs:restriction base="xs:string">
36         <xs:enumeration value="both" />
37         <xs:enumeration value="negative" />
38         <xs:enumeration value="positive" />
39         <xs:enumeration value="unknown" />
40     </xs:restriction>
41 </xs:simpleType>
42 <xs:complexType name="AlertCLinear" abstract="true">
43     <xs:sequence>
44         <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
    ↪ minOccurs="1" maxOccurs="1" />
45         <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
    ↪ minOccurs="1" maxOccurs="1" />
46         <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
    ↪ minOccurs="1" maxOccurs="1" />
47         <xs:element name="alertCLinearExtension" type="D2LogicalModel:_ExtensionType"
    ↪ minOccurs="0" />
48     </xs:sequence>
49 </xs:complexType>
50 <xs:complexType name="AlertCLinearByCode">
51     <xs:complexContent>
52         <xs:extension base="D2LogicalModel:AlertCLinear">
53             <xs:sequence>
54                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
55                 <xs:element name="locationCodeForLinearLocation" type=
    ↪ "D2LogicalModel:AlertCLocation" />
56                 <xs:element name="alertCLinearByCodeExtension" type="D2LogicalModel:_
    ↪ ExtensionType" minOccurs="0" />
57             </xs:sequence>
58         </xs:extension>
59     </xs:complexContent>
60 </xs:complexType>
61 <xs:complexType name="AlertCLocation">
62     <xs:sequence>
63         <xs:element name="alertCLocationName" type="D2LogicalModel:MultilingualString"
    ↪ minOccurs="0" maxOccurs="1" />
64         <xs:element name="specificLocation" type="D2LogicalModel:AlertCLocationCode"
    ↪ minOccurs="1" maxOccurs="1" />
65         <xs:element name="alertCLocationExtension" type="D2LogicalModel:_ExtensionType"
    ↪ minOccurs="0" />
66     </xs:sequence>
67 </xs:complexType>
68 <xs:simpleType name="AlertCLocationCode">
69     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
70 </xs:simpleType>
71 <xs:complexType name="AlertCMethod2Linear">
72     <xs:complexContent>
73         <xs:extension base="D2LogicalModel:AlertCLinear">
74             <xs:sequence>

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```

75     <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
76     <xs:element name="alertCMethod2PrimaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
77     <xs:element name="alertCMethod2SecondaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2SecondaryPointLocation" />
78     <xs:element name="alertCMethod2LinearExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
79     </xs:sequence>
80     </xs:extension>
81     </xs:complexContent>
82 </xs:complexType>
83 <xs:complexType name="AlertCMethod2Point">
84     <xs:complexContent>
85         <xs:extension base="D2LogicalModel:AlertCPoint">
86             <xs:sequence>
87                 <xs:element name="alertCDirection" type="D2LogicalModel:AlertCDirection" />
88                 <xs:element name="alertCMethod2PrimaryPointLocation" type=
↪ "D2LogicalModel:AlertCMethod2PrimaryPointLocation" />
89                 <xs:element name="alertCMethod2PointExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
90             </xs:sequence>
91             </xs:extension>
92         </xs:complexContent>
93     </xs:complexType>
94 <xs:complexType name="AlertCMethod2PrimaryPointLocation">
95     <xs:sequence>
96         <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
97         <xs:element name="alertCMethod2PrimaryPointLocationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
98     </xs:sequence>
99 </xs:complexType>
100 <xs:complexType name="AlertCMethod2SecondaryPointLocation">
101     <xs:sequence>
102         <xs:element name="alertCLocation" type="D2LogicalModel:AlertCLocation" />
103         <xs:element name="alertCMethod2SecondaryPointLocationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
104     </xs:sequence>
105 </xs:complexType>
106 <xs:complexType name="AlertCPoint" abstract="true">
107     <xs:sequence>
108         <xs:element name="alertCLocationCountryCode" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
109         <xs:element name="alertCLocationTableNumber" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
110         <xs:element name="alertCLocationTableVersion" type="D2LogicalModel:String"
↪ minOccurs="1" maxOccurs="1" />
111         <xs:element name="alertCPointExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
112     </xs:sequence>
113 </xs:complexType>
114 <xs:simpleType name="AngleInDegrees">
115     <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
116 </xs:simpleType>
117 <xs:simpleType name="AreaOfInterestEnum">
118     <xs:restriction base="xs:string">
119         <xs:enumeration value="continentWide" />

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```

120     <xs:enumeration value="national" />
121     <xs:enumeration value="neighbouringCountries" />
122     <xs:enumeration value="notSpecified" />
123     <xs:enumeration value="regional" />
124   </xs:restriction>
125 </xs:simpleType>
126 <xs:simpleType name="Boolean">
127   <xs:restriction base="xs:boolean" />
128 </xs:simpleType>
129 <xs:simpleType name="ConfidentialityValueEnum">
130   <xs:restriction base="xs:string">
131     <xs:enumeration value="internalUse" />
132     <xs:enumeration value="noRestriction" />
133     <xs:enumeration value="restrictedToAuthorities" />
134     <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators" />
135     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers" />
136     <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms" />
137   </xs:restriction>
138 </xs:simpleType>
139 <xs:simpleType name="CountryEnum">
140   <xs:restriction base="xs:string">
141     <xs:enumeration value="at" />
142     <xs:enumeration value="be" />
143     <xs:enumeration value="bg" />
144     <xs:enumeration value="ch" />
145     <xs:enumeration value="cs" />
146     <xs:enumeration value="cy" />
147     <xs:enumeration value="cz" />
148     <xs:enumeration value="de" />
149     <xs:enumeration value="dk" />
150     <xs:enumeration value="ee" />
151     <xs:enumeration value="es" />
152     <xs:enumeration value="fi" />
153     <xs:enumeration value="fo" />
154     <xs:enumeration value="fr" />
155     <xs:enumeration value="gb" />
156     <xs:enumeration value="gg" />
157     <xs:enumeration value="gi" />
158     <xs:enumeration value="gr" />
159     <xs:enumeration value="hr" />
160     <xs:enumeration value="hu" />
161     <xs:enumeration value="ie" />
162     <xs:enumeration value="im" />
163     <xs:enumeration value="is" />
164     <xs:enumeration value="it" />
165     <xs:enumeration value="je" />
166     <xs:enumeration value="li" />
167     <xs:enumeration value="lt" />
168     <xs:enumeration value="lu" />
169     <xs:enumeration value="lv" />
170     <xs:enumeration value="ma" />
171     <xs:enumeration value="mc" />
172     <xs:enumeration value="mk" />
173     <xs:enumeration value="mt" />
174     <xs:enumeration value="nl" />
175     <xs:enumeration value="no" />

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```

176     <xs:enumeration value="pl" />
177     <xs:enumeration value="pt" />
178     <xs:enumeration value="ro" />
179     <xs:enumeration value="se" />
180     <xs:enumeration value="si" />
181     <xs:enumeration value="sk" />
182     <xs:enumeration value="sm" />
183     <xs:enumeration value="tr" />
184     <xs:enumeration value="va" />
185     <xs:enumeration value="other" />
186   </xs:restriction>
187 </xs:simpleType>
188 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
189   <xs:unique name="_d2LogicalModelPredefinedLocationConstraint">
190     <xs:selector xpath="."/><xs:field xpath="@id" />
191     <xs:field xpath="@version" />
192   </xs:unique>
193 </xs:element>
194 <xs:complexType name="D2LogicalModel">
195   <xs:sequence>
196     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
197     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication"
198     ↪ minOccurs="0" />
199     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType"
200     ↪ minOccurs="0" />
201   </xs:sequence>
202   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
203 </xs:complexType>
204 <xs:simpleType name="DateTime">
205   <xs:restriction base="xs:dateTime" />
206 </xs:simpleType>
207 <xs:simpleType name="DirectionEnum">
208   <xs:restriction base="xs:string">
209     <xs:enumeration value="allDirections" />
210     <xs:enumeration value="bothWays" />
211     <xs:enumeration value="clockwise" />
212     <xs:enumeration value="anticlockwise" />
213     <xs:enumeration value="innerRing" />
214     <xs:enumeration value="outerRing" />
215     <xs:enumeration value="northBound" />
216     <xs:enumeration value="northEastBound" />
217     <xs:enumeration value="eastBound" />
218     <xs:enumeration value="southEastBound" />
219     <xs:enumeration value="southBound" />
220     <xs:enumeration value="southWestBound" />
221     <xs:enumeration value="westBound" />
222     <xs:enumeration value="northWestBound" />
223     <xs:enumeration value="inboundTowardsTown" />
224     <xs:enumeration value="outboundFromTown" />
225     <xs:enumeration value="unknown" />
226     <xs:enumeration value="opposite" />
227     <xs:enumeration value="other" />
228   </xs:restriction>
229 </xs:simpleType>
230 <xs:complexType name="DistanceAlongLinearElement" abstract="true">

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```

230     <xs:sequence>
231       <xs:element name="distanceAlongLinearElementExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
232     </xs:sequence>
233   </xs:complexType>
234   <xs:complexType name="DistanceFromLinearElementReferent">
235     <xs:complexContent>
236       <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
237         <xs:sequence>
238           <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪minOccurs="1" maxOccurs="1" />
239           <xs:element name="fromReferent" type="D2LogicalModel:Referent" />
240           <xs:element name="towardsReferent" type="D2LogicalModel:Referent" minOccurs=
↪"0" />
241           <xs:element name="distanceFromLinearElementReferentExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
242         </xs:sequence>
243       </xs:extension>
244     </xs:complexContent>
245   </xs:complexType>
246   <xs:complexType name="DistanceFromLinearElementStart">
247     <xs:complexContent>
248       <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
249         <xs:sequence>
250           <xs:element name="distanceAlong" type="D2LogicalModel:MetresAsFloat" ↪
↪minOccurs="1" maxOccurs="1" />
251           <xs:element name="distanceFromLinearElementStartExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
252         </xs:sequence>
253       </xs:extension>
254     </xs:complexContent>
255   </xs:complexType>
256   <xs:complexType name="Exchange">
257     <xs:sequence>
258       <xs:element name="supplierIdentification" type=
↪"D2LogicalModel:InternationalIdentifier" />
259       <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType" ↪
↪minOccurs="0" />
260     </xs:sequence>
261   </xs:complexType>
262   <xs:simpleType name="Float">
263     <xs:restriction base="xs:float" />
264   </xs:simpleType>
265   <xs:simpleType name="GeometryTypeEnum">
266     <xs:restriction base="xs:string">
267       <xs:enumeration value="continuous" />
268       <xs:enumeration value="discontinuous" />
269     </xs:restriction>
270   </xs:simpleType>
271   <xs:complexType name="GlobalNetworkLinear">
272     <xs:sequence>
273       <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString" ↪
↪minOccurs="1" maxOccurs="1" />
274       <xs:element name="linearGeometryType" type="D2LogicalModel:GeometryTypeEnum" ↪
↪minOccurs="1" maxOccurs="1" />
275       <xs:element name="startPoint" type="D2LogicalModel:PointByMultiCoordinates" ↪

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```

276   <minOccurs="0" />
      <xs:element name="endPoint" type="D2LogicalModel:PointByMultiCoordinates"
277   <minOccurs="0" />
      <xs:element name="linearWithinLinearGNElement" type=
278   <"D2LogicalModel:LinearWithinLinearGNElement" maxOccurs="unbounded" />
      </xs:sequence>
279   </xs:complexType>
280   <xs:complexType name="GroupOfLocations" abstract="true">
281     <xs:sequence>
282       <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType"
283       <" minOccurs="0" />
      </xs:sequence>
284   </xs:complexType>
285   <xs:complexType name="HeaderInformation">
286     <xs:sequence>
287       <xs:element name="areaOfInterest" type="D2LogicalModel:AreaOfInterestEnum"
288       <minOccurs="0" maxOccurs="1" />
      <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum"
289       <" minOccurs="1" maxOccurs="1" />
      <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum"
290       <" minOccurs="1" maxOccurs="1" />
      <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
291       <maxOccurs="1" />
      <xs:element name="headerInformationExtension" type="D2LogicalModel:_
292       <ExtensionType" minOccurs="0" />
      </xs:sequence>
293   </xs:complexType>
294   <xs:simpleType name="HeightGradeEnum">
295     <xs:restriction base="xs:string">
296       <xs:enumeration value="aboveGrade" />
297       <xs:enumeration value="atGrade" />
298       <xs:enumeration value="belowGrade" />
299     </xs:restriction>
300   </xs:simpleType>
301   <xs:simpleType name="InformationStatusEnum">
302     <xs:restriction base="xs:string">
303       <xs:enumeration value="real" />
304       <xs:enumeration value="securityExercise" />
305       <xs:enumeration value="technicalExercise" />
306       <xs:enumeration value="test" />
307     </xs:restriction>
308   </xs:simpleType>
309   <xs:complexType name="InternationalIdentifier">
310     <xs:sequence>
311       <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
312       <maxOccurs="1" />
      <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1"
313       <" maxOccurs="1" />
      <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
314       <ExtensionType" minOccurs="0" />
      </xs:sequence>
315   </xs:complexType>
316   <xs:simpleType name="Language">
317     <xs:restriction base="xs:language" />
318   </xs:simpleType>
319   <xs:complexType name="Linear">

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```

320     <xs:complexContent>
321       <xs:extension base="D2LogicalModel:NetworkLocation">
322         <xs:sequence>
323           <xs:element name="alertCLinear" type="D2LogicalModel:AlertCLinear"
324             ↪ minOccurs="0" />
325           <xs:element name="linearWithinLinearElement" type=
326             ↪ "D2LogicalModel:LinearWithinLinearElement" minOccurs="0" />
327           <xs:element name="globalNetworkLinear" type=
328             ↪ "D2LogicalModel:GlobalNetworkLinear" minOccurs="0" />
329           <xs:element name="linearExtension" type="D2LogicalModel:_LinearExtensionType"
330             ↪ minOccurs="0" />
331         </xs:sequence>
332       </xs:extension>
333     </xs:complexContent>
334   </xs:complexType>
335   <xs:complexType name="LinearElement">
336     <xs:sequence>
337       <xs:element name="roadName" type="D2LogicalModel:MultilingualString" minOccurs=
338         ↪ "0" maxOccurs="1" />
339       <xs:element name="roadNumber" type="D2LogicalModel:String" minOccurs="0"
340         ↪ maxOccurs="1" />
341       <xs:element name="linearElementReferenceModel" type="D2LogicalModel:String"
342         ↪ minOccurs="0" maxOccurs="1" />
343       <xs:element name="linearElementReferenceModelVersion" type=
344         ↪ "D2LogicalModel:String" minOccurs="0" maxOccurs="1" />
345       <xs:element name="linearElementNature" type=
346         ↪ "D2LogicalModel:LinearElementNatureEnum" minOccurs="0" maxOccurs="1" />
347       <xs:element name="linearElementExtension" type="D2LogicalModel:_ExtensionType"
348         ↪ minOccurs="0" />
349     </xs:sequence>
350   </xs:complexType>
351   <xs:complexType name="LinearElementByCode">
352     <xs:complexContent>
353       <xs:extension base="D2LogicalModel:LinearElement">
354         <xs:sequence>
355           <xs:element name="linearElementIdentifier" type="D2LogicalModel:String"
356             ↪ minOccurs="1" maxOccurs="1" />
357           <xs:element name="linearElementByCodeExtension" type="D2LogicalModel:_
358             ↪ ExtensionType" minOccurs="0" />
359         </xs:sequence>
360       </xs:extension>
361     </xs:complexContent>
362   </xs:complexType>
363   <xs:complexType name="LinearElementByPoints">
364     <xs:complexContent>
365       <xs:extension base="D2LogicalModel:LinearElement">
366         <xs:sequence>
367           <xs:element name="startPointOfLinearElement" type="D2LogicalModel:Referent"
368             ↪ />
369           <xs:element name="intermediatePointOnLinearElement" type="D2LogicalModel:_
370             ↪ IntermediatePointOnLinearElement" minOccurs="0" maxOccurs="unbounded" />
371           <xs:element name="endPointOfLinearElement" type="D2LogicalModel:Referent" />
372           <xs:element name="linearElementByPointsExtension" type="D2LogicalModel:_
373             ↪ ExtensionType" minOccurs="0" />
374         </xs:sequence>
375       </xs:extension>
376     </xs:complexContent>
377   </xs:complexType>

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```

361     </xs:complexContent>
362 </xs:complexType>
363 <xs:simpleType name="LinearElementNatureEnum">
364     <xs:restriction base="xs:string">
365         <xs:enumeration value="road" />
366         <xs:enumeration value="roadSection" />
367         <xs:enumeration value="slipRoad" />
368         <xs:enumeration value="other" />
369     </xs:restriction>
370 </xs:simpleType>
371 <xs:simpleType name="LinearReferencingDirectionEnum">
372     <xs:restriction base="xs:string">
373         <xs:enumeration value="both" />
374         <xs:enumeration value="opposite" />
375         <xs:enumeration value="aligned" />
376         <xs:enumeration value="unknown" />
377     </xs:restriction>
378 </xs:simpleType>
379 <xs:complexType name="LinearWithinLinearElement">
380     <xs:sequence>
381         <xs:element name="administrativeAreaOfLinearSection" type=
382         ↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
383         <xs:element name="directionBoundOnLinearSection" type=
384         ↪ "D2LogicalModel:DirectionEnum" minOccurs="0" maxOccurs="1" />
385         <xs:element name="directionRelativeOnLinearSection" type=
386         ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="0" maxOccurs="1" />
387         <xs:element name="heightGradeOfLinearSection" type=
388         ↪ "D2LogicalModel:HeightGradeEnum" minOccurs="0" maxOccurs="1" />
389         <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
390         <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
391         <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
392         <xs:element name="linearWithinLinearElementExtension" type="D2LogicalModel:_
393         ↪ ExtensionType" minOccurs="0" />
394     </xs:sequence>
395 </xs:complexType>
396 <xs:complexType name="LinearWithinLinearGNElement">
397     <xs:sequence>
398         <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
399         ↪ "0" maxOccurs="1" />
400         <xs:element name="directionRelativeOnLinearSection" type=
401         ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
402         <xs:element name="orderOfSection" type="D2LogicalModel:NonNegativeInteger"
403         ↪ minOccurs="0" maxOccurs="1" />
404         <xs:element name="fromPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
405         <xs:element name="toPoint" type="D2LogicalModel:DistanceAlongLinearElement" />
406         <xs:element name="linearWithinLinearGNElementExtension" type="D2LogicalModel:_
407         ↪ ExtensionType" minOccurs="0" />
408     </xs:sequence>
409 </xs:complexType>
410 <xs:complexType name="Location" abstract="true">
411     <xs:complexContent>
412         <xs:extension base="D2LogicalModel:GroupOfLocations">
413             <xs:sequence>
414                 <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
415                 ↪ minOccurs="0" />
416             </xs:sequence>

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407     </xs:extension>
408   </xs:complexContent>
409 </xs:complexType>
410 <xs:simpleType name="MetresAsFloat">
411   <xs:restriction base="D2LogicalModel:Float" />
412 </xs:simpleType>
413 <xs:simpleType name="MetresAsNonNegativeInteger">
414   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
415 </xs:simpleType>
416 <xs:complexType name="MultilingualString">
417   <xs:sequence>
418     <xs:element name="values">
419       <xs:complexType>
420         <xs:sequence>
421           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
↳ maxOccurs="unbounded" />
422         </xs:sequence>
423       </xs:complexType>
424     </xs:element>
425   </xs:sequence>
426 </xs:complexType>
427 <xs:complexType name="MultilingualStringValue">
428   <xs:simpleContent>
429     <xs:extension base="D2LogicalModel:MultilingualStringValueType">
430       <xs:attribute name="lang" type="xs:language" />
431     </xs:extension>
432   </xs:simpleContent>
433 </xs:complexType>
434 <xs:simpleType name="MultilingualStringValueType">
435   <xs:restriction base="xs:string">
436     <xs:maxLength value="1024" />
437   </xs:restriction>
438 </xs:simpleType>
439 <xs:complexType name="NetworkLocation" abstract="true">
440   <xs:complexContent>
441     <xs:extension base="D2LogicalModel:Location">
442       <xs:sequence>
443         <xs:element name="networkLocationExtension" type="D2LogicalModel:_
↳ ExtensionType" minOccurs="0" />
444       </xs:sequence>
445     </xs:extension>
446   </xs:complexContent>
447 </xs:complexType>
448 <xs:simpleType name="NonNegativeInteger">
449   <xs:restriction base="xs:nonNegativeInteger" />
450 </xs:simpleType>
451 <xs:complexType name="OpenlrBaseLocationReferencePoint" abstract="true">
452   <xs:sequence>
453     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
454     <xs:element name="openlrLineAttributes" type=
↳ "D2LogicalModel:OpenlrLineAttributes" />
455     <xs:element name="openlrBaseLocationReferencePointExtension" type=
↳ "D2LogicalModel:_ExtensionType" minOccurs="0" />
456   </xs:sequence>
457 </xs:complexType>
458 <xs:complexType name="OpenlrBasePointLocation" abstract="true">

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459     <xs:sequence>
460       <xs:element name="openlrSideOfRoad" type="D2LogicalModel:OpenlrSideOfRoadEnum"
461         ↪ minOccurs="1" maxOccurs="1" />
462       <xs:element name="openlrOrientation" type="D2LogicalModel:OpenlrOrientationEnum"
463         ↪ minOccurs="1" maxOccurs="1" />
464       <xs:element name="openlrPositiveOffset" type=
465         ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
466       <xs:element name="openlrLocationReferencePoint" type=
467         ↪ "D2LogicalModel:OpenlrLocationReferencePoint" />
468       <xs:element name="openlrLastLocationReferencePoint" type=
469         ↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
470       <xs:element name="openlrBasePointLocationExtension" type="D2LogicalModel:_
471         ↪ ExtensionType" minOccurs="0" />
472     </xs:sequence>
473   </xs:complexType>
474   <xs:complexType name="OpenlrExtendedLinear">
475     <xs:sequence>
476       <xs:element name="firstDirection" type=
477         ↪ "D2LogicalModel:OpenlrLineLocationReference" />
478       <xs:element name="oppositeDirection" type=
479         ↪ "D2LogicalModel:OpenlrLineLocationReference" minOccurs="0" />
480     </xs:sequence>
481   </xs:complexType>
482   <xs:complexType name="OpenlrExtendedPoint">
483     <xs:sequence>
484       <xs:element name="openlrPointLocationReference" type=
485         ↪ "D2LogicalModel:OpenlrPointLocationReference" />
486     </xs:sequence>
487   </xs:complexType>
488   <xs:simpleType name="OpenlrFormOfWayEnum">
489     <xs:restriction base="xs:string">
490       <xs:enumeration value="undefined" />
491       <xs:enumeration value="motorway" />
492       <xs:enumeration value="multipleCarriageway" />
493       <xs:enumeration value="singleCarriageway" />
494       <xs:enumeration value="roundabout" />
495       <xs:enumeration value="slipRoad" />
496       <xs:enumeration value="trafficSquare" />
497       <xs:enumeration value="other" />
498     </xs:restriction>
499   </xs:simpleType>
500   <xs:simpleType name="OpenlrFunctionalRoadClassEnum">
501     <xs:restriction base="xs:string">
502       <xs:enumeration value="FRC0" />
503       <xs:enumeration value="FRC1" />
504       <xs:enumeration value="FRC2" />
505       <xs:enumeration value="FRC3" />
506       <xs:enumeration value="FRC4" />
507       <xs:enumeration value="FRC5" />
508       <xs:enumeration value="FRC6" />
509       <xs:enumeration value="FRC7" />

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501   </xs:restriction>
502 </xs:simpleType>
503 <xs:complexType name="OpenlrGeoCoordinate">
504   <xs:sequence>

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505     <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" />
506     <xs:element name="openlrGeoCoordinateExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
507   </xs:sequence>
508 </xs:complexType>
509 <xs:complexType name="OpenlrLastLocationReferencePoint">
510   <xs:complexContent>
511     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
512       <xs:sequence>
513         <xs:element name="openlrLastLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
514       </xs:sequence>
515     </xs:extension>
516   </xs:complexContent>
517 </xs:complexType>
518 <xs:complexType name="OpenlrLineAttributes">
519   <xs:sequence>
520     <xs:element name="openlrFunctionalRoadClass" type=
↪ "D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
521     <xs:element name="openlrFormOfWay" type="D2LogicalModel:OpenlrFormOfWayEnum"
↪ minOccurs="1" maxOccurs="1" />
522     <xs:element name="openlrBearing" type="D2LogicalModel:AngleInDegrees" minOccurs=
↪ "1" maxOccurs="1" />
523     <xs:element name="openlrLineAttributesExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
524   </xs:sequence>
525 </xs:complexType>
526 <xs:complexType name="OpenlrLineLocationReference">
527   <xs:sequence>
528     <xs:element name="openlrLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLocationReferencePoint" maxOccurs="unbounded" />
529     <xs:element name="openlrLastLocationReferencePoint" type=
↪ "D2LogicalModel:OpenlrLastLocationReferencePoint" />
530     <xs:element name="openlrOffsets" type="D2LogicalModel:OpenlrOffsets" minOccurs=
↪ "0" />
531     <xs:element name="openlrLineLocationReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
532   </xs:sequence>
533 </xs:complexType>
534 <xs:complexType name="OpenlrLocationReferencePoint">
535   <xs:complexContent>
536     <xs:extension base="D2LogicalModel:OpenlrBaseLocationReferencePoint">
537       <xs:sequence>
538         <xs:element name="openlrPathAttributes" type=
↪ "D2LogicalModel:OpenlrPathAttributes" />
539         <xs:element name="openlrLocationReferencePointExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
540       </xs:sequence>
541     </xs:extension>
542   </xs:complexContent>
543 </xs:complexType>
544 <xs:complexType name="OpenlrOffsets">
545   <xs:sequence>
546     <xs:element name="openlrPositiveOffset" type=
↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
547     <xs:element name="openlrNegativeOffset" type=

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548   <D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1" />
      <xs:element name="openlrOffsetsExtension" type="D2LogicalModel:_ExtensionType"
549   <minOccurs="0" />
      </xs:sequence>
550   </xs:complexType>
551   <xs:simpleType name="OpenlrOrientationEnum">
552     <xs:restriction base="xs:string">
553       <xs:enumeration value="noOrientationOrUnknown" />
554       <xs:enumeration value="withLineDirection" />
555       <xs:enumeration value="againstLineDirection" />
556       <xs:enumeration value="both" />
557     </xs:restriction>
558   </xs:simpleType>
559   <xs:complexType name="OpenlrPathAttributes">
560     <xs:sequence>
561       <xs:element name="openlrLowestFRCToNextLRPoint" type=
562   <D2LogicalModel:OpenlrFunctionalRoadClassEnum" minOccurs="1" maxOccurs="1" />
       <xs:element name="openlrDistanceToNextLRPoint" type=
563   <D2LogicalModel:NonNegativeInteger" minOccurs="1" maxOccurs="1" />
       <xs:element name="openlrPathAttributesExtension" type="D2LogicalModel:_
564   <ExtensionType" minOccurs="0" />
       </xs:sequence>
565     </xs:complexType>
566   <xs:complexType name="OpenlrPointAlongLine">
567     <xs:complexContent>
568       <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
569         <xs:sequence>
570           <xs:element name="openlrPointAlongLineExtension" type="D2LogicalModel:_
571   <ExtensionType" minOccurs="0" />
           </xs:sequence>
572         </xs:extension>
573       </xs:complexContent>
574     </xs:complexType>
575   <xs:complexType name="OpenlrPointLocationReference">
576     <xs:sequence>
577       <xs:element name="openlrGeoCoordinate" type="D2LogicalModel:OpenlrGeoCoordinate
578   <" minOccurs="0" />
       <xs:element name="openlrPoiWithAccessPoint" type=
579   <D2LogicalModel:OpenlrPoiWithAccessPoint" minOccurs="0" />
       <xs:element name="openlrPointAlongLine" type=
580   <D2LogicalModel:OpenlrPointAlongLine" minOccurs="0" />
       <xs:element name="openlrPointLocationReferenceExtension" type="D2LogicalModel:_
581   <ExtensionType" minOccurs="0" />
       </xs:sequence>
582     </xs:complexType>
583   <xs:complexType name="OpenlrPoiWithAccessPoint">
584     <xs:complexContent>
585       <xs:extension base="D2LogicalModel:OpenlrBasePointLocation">
586         <xs:sequence>
587           <xs:element name="openlrCoordinate" type="D2LogicalModel:PointCoordinates" /
588   <
           <xs:element name="openlrPoiWithAccessPointExtension" type="D2LogicalModel:_
589   <ExtensionType" minOccurs="0" />
           </xs:sequence>
590         </xs:extension>
591       </xs:complexContent>

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592 </xs:complexType>
593 <xs:simpleType name="OpenlrSideOfRoadEnum">
594   <xs:restriction base="xs:string">
595     <xs:enumeration value="onRoadOrUnknown" />
596     <xs:enumeration value="right" />
597     <xs:enumeration value="left" />
598     <xs:enumeration value="both" />
599   </xs:restriction>
600 </xs:simpleType>
601 <xs:complexType name="PayloadPublication" abstract="true">
602   <xs:sequence>
603     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
604     ↪ maxOccurs="1" />
605     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
606     ↪ maxOccurs="1" />
607     <xs:element name="publicationCreator" type=
608     ↪ "D2LogicalModel:InternationalIdentifier" />
609     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
610     ↪ ExtensionType" minOccurs="0" />
611   </xs:sequence>
612   <xs:attribute name="lang" type="D2LogicalModel:Language" use="required" />
613 </xs:complexType>
614 <xs:simpleType name="Percentage">
615   <xs:restriction base="D2LogicalModel:Float" />
616 </xs:simpleType>
617 <xs:complexType name="PercentageDistanceAlongLinearElement">
618   <xs:complexContent>
619     <xs:extension base="D2LogicalModel:DistanceAlongLinearElement">
620       <xs:sequence>
621         <xs:element name="percentageDistanceAlong" type="D2LogicalModel:Percentage"
622         ↪ minOccurs="1" maxOccurs="1" />
623         <xs:element name="percentageDistanceAlongLinearElementExtension" type=
624         ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
625       </xs:sequence>
626     </xs:extension>
627   </xs:complexContent>
628 </xs:complexType>
629 <xs:complexType name="Point">
630   <xs:complexContent>
631     <xs:extension base="D2LogicalModel:NetworkLocation">
632       <xs:sequence>
633         <xs:element name="alertCPoint" type="D2LogicalModel:AlertCPoint" minOccurs=
634         ↪ "0" />
635         <xs:element name="pointAlongLinearElement" type=
636         ↪ "D2LogicalModel:PointAlongLinearElement" minOccurs="0" />
637         <xs:element name="pointByCoordinates" type=
638         ↪ "D2LogicalModel:PointByCoordinates" minOccurs="0" />
639         <xs:element name="pointWithinLinearGNElement" type=
640         ↪ "D2LogicalModel:PointWithinLinearGNElement" minOccurs="0" />
641         <xs:element name="pointByMultiCoordinates" type=
642         ↪ "D2LogicalModel:PointByMultiCoordinates" minOccurs="0" />
643         <xs:element name="pointExtension" type="D2LogicalModel:_PointExtensionType"
644         ↪ minOccurs="0" />
645       </xs:sequence>
646     </xs:extension>
647   </xs:complexContent>

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636 </xs:complexType>
637 <xs:complexType name="PointAlongLinearElement">
638   <xs:sequence>
639     <xs:element name="linearElement" type="D2LogicalModel:LinearElement" />
640     <xs:element name="distanceAlongLinearElement" type=
641     ↪ "D2LogicalModel:DistanceAlongLinearElement" />
642     <xs:element name="pointAlongLinearElementExtension" type="D2LogicalModel:_
643     ↪ ExtensionType" minOccurs="0" />
644   </xs:sequence>
645 </xs:complexType>
646 <xs:complexType name="PointByCoordinates">
647   <xs:sequence>
648     <xs:element name="bearing" type="D2LogicalModel:NonNegativeInteger" minOccurs="0
649     ↪ " maxOccurs="1" />
650     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates" />
651     <xs:element name="pointByCoordinatesExtension" type="D2LogicalModel:_
652     ↪ ExtensionType" minOccurs="0" />
653   </xs:sequence>
654 </xs:complexType>
655 <xs:complexType name="PointByMultiCoordinates">
656   <xs:sequence>
657     <xs:element name="sjtskPointCoordinates" type=
658     ↪ "D2LogicalModel:SJTSPPointCoordinates" minOccurs="0" />
659     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
660     ↪ minOccurs="0" />
661     <xs:element name="pointByMultiCoordinatesExtension" type="D2LogicalModel:_
662     ↪ ExtensionType" minOccurs="0" />
663   </xs:sequence>
664 </xs:complexType>
665 <xs:complexType name="PointCoordinates">
666   <xs:sequence>
667     <xs:element name="latitude" type="D2LogicalModel:Float" minOccurs="1" maxOccurs=
668     ↪ "1" />
669     <xs:element name="longitude" type="D2LogicalModel:Float" minOccurs="1"
670     ↪ maxOccurs="1" />
671     <xs:element name="pointCoordinatesExtension" type="D2LogicalModel:_ExtensionType
672     ↪ " minOccurs="0" />
673   </xs:sequence>
674 </xs:complexType>
675 <xs:complexType name="PointWithinLinearGNElement">
676   <xs:sequence>
677     <xs:element name="sectionId" type="D2LogicalModel:NonNegativeInteger" minOccurs=
678     ↪ "0" maxOccurs="1" />
679     <xs:element name="directionRelativeOnLinearSection" type=
680     ↪ "D2LogicalModel:LinearReferencingDirectionEnum" minOccurs="1" maxOccurs="1" />
681     <xs:element name="networkVersion" type="D2LogicalModel:MultilingualString"
682     ↪ minOccurs="1" maxOccurs="1" />
683     <xs:element name="distanceAlongLinearElement" type=
684     ↪ "D2LogicalModel:DistanceAlongLinearElement" />
685     <xs:element name="pointWithinLinearGNElementExtension" type="D2LogicalModel:_
686     ↪ ExtensionType" minOccurs="0" />
687   </xs:sequence>
688 </xs:complexType>
689 <xs:complexType name="PredefinedLocation">
690   <xs:complexContent>
691     <xs:extension base="D2LogicalModel:PredefinedLocationContainer">

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677     <xs:sequence>
678       <xs:element name="predefinedLocationName" type=
↪ "D2LogicalModel:MultilingualString" minOccurs="0" maxOccurs="1" />
679       <xs:element name="location" type="D2LogicalModel:Location" />
680       <xs:element name="predefinedLocationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
681     </xs:sequence>
682     <xs:attribute name="id" type="xs:string" use="required" />
683     <xs:attribute name="version" type="xs:string" use="required" />
684   </xs:extension>
685 </xs:complexType>
686 </xs:complexType>
687 <xs:complexType name="PredefinedLocationContainer" abstract="true">
688   <xs:sequence>
689     <xs:element name="predefinedLocationContainerExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
690   </xs:sequence>
691 </xs:complexType>
692 <xs:complexType name="PredefinedLocationsPublication">
693   <xs:complexContent>
694     <xs:extension base="D2LogicalModel:PayloadPublication">
695       <xs:sequence>
696         <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation
↪ " />
697         <xs:element name="predefinedLocationContainer" type=
↪ "D2LogicalModel:PredefinedLocationContainer" maxOccurs="unbounded" />
698         <xs:element name="predefinedLocationsPublicationExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
699       </xs:sequence>
700     </xs:extension>
701   </xs:complexContent>
702 </xs:complexType>
703 <xs:complexType name="Referent">
704   <xs:sequence>
705     <xs:element name="referentIdentifier" type="D2LogicalModel:String" minOccurs="1
↪ " maxOccurs="1" />
706     <xs:element name="referentName" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1" />
707     <xs:element name="referentType" type="D2LogicalModel:ReferentTypeEnum"
↪ minOccurs="1" maxOccurs="1" />
708     <xs:element name="referentDescription" type="D2LogicalModel:MultilingualString"
↪ minOccurs="0" maxOccurs="1" />
709     <xs:element name="pointCoordinates" type="D2LogicalModel:PointCoordinates"
↪ minOccurs="0" />
710     <xs:element name="referentExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
711   </xs:sequence>
712 </xs:complexType>
713 <xs:simpleType name="ReferentTypeEnum">
714   <xs:restriction base="xs:string">
715     <xs:enumeration value="boundary" />
716     <xs:enumeration value="intersection" />
717     <xs:enumeration value="referenceMarker" />
718     <xs:enumeration value="landmark" />
719     <xs:enumeration value="roadNode" />
720   </xs:restriction>

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721 </xs:simpleType>
722 <xs:complexType name="SJTSKPointCoordinates">
723   <xs:sequence>
724     <xs:element name="sjtskX" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
725     ↪"/>
726     <xs:element name="sjtskY" type="D2LogicalModel:Float" minOccurs="1" maxOccurs="1"
727     ↪"/>
728     <xs:element name="sjtskPointCoordinatesExtension" type="D2LogicalModel:_
729     ↪ExtensionType" minOccurs="0" />
730   </xs:sequence>
731 </xs:complexType>
732 <xs:simpleType name="String">
733   <xs:restriction base="xs:string">
734     <xs:maxLength value="1024" />
735   </xs:restriction>
736 </xs:simpleType>
737 <xs:simpleType name="UrgencyEnum">
738   <xs:restriction base="xs:string">
739     <xs:enumeration value="extremelyUrgent" />
740     <xs:enumeration value="urgent" />
741     <xs:enumeration value="normalUrgency" />
742   </xs:restriction>
743 </xs:simpleType>
744 </xs:schema>

```

CHANGELOG

version 1.1.0

- added pointByCoordinates to concepts, schema and samples
- added feed type to schema and samples
- added OpenLR location to schema

version 1.0.1

- revision of the documentation, proofreading, changes to examples

version 1.0.0

- Initial format and documentation

BIBLIOGRAPHY

- [16157-1] CEN TS 16157-1: Intelligent transport systems – DATEX II data exchange specifications for traffic management and information – Part 1: Context and framework
- [16157-2] CEN TS 16157-2: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 2: Location referencing
- [16157-3] CEN TS 16157-3: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 3: Situation Publication
- [16157-4] CEN TS 16157-4: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 4: Variable Message Sign (VMS) Publications
- [16157-5] CEN TS 16157-5: Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 5: Measured and elaborated data publications
- [d2ref] DATEX II Current version/Reference set, <https://docs.datex2.eu/downloads/>.
- [d2supp] DATEX II Current version/Supporting, <https://docs.datex2.eu/>.

Symbols

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(DATEX II) profile, [3](#)
(W3C) XML schema, [3](#)

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B level extension, [3](#)

C

C level extension, [3](#)

D

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